

Training Report
Multi-Agent Systems, Social Sciences
and Integrated Natural Resource Management: Methodological Issues
Faculty of Science, Chulalongkorn University, Bangkok, Thailand
17-21 March 2003

Key resource person

Pr. Scott Moss, Director,
Center for Policy Modelling, Manchester Metropolitan University (CPM-MMU), UK.

IRRI Staff participants

François Bousquet and Guy Trébuil.

Local organizers

- Dr. Nantana Gajasen, Department of Biology, Faculty of Science, Chulalongkorn University, Bangkok.
- Wipawan Boonsom, secretary, IRRI-Cirad Project at IRRI Thailand Office, Bangkok.
- The course was opened by Dr. Nuanthip Kamolvarin, Assistant to the President for Research Affairs, Chulalongkorn University. Pr. Siriwat Wongsiri, Head of Department of Biology at the Faculty of Science, Chulalongkorn University, presided over the farewell dinner and the closing ceremony.

Group photo



Purpose

This one-week course corresponded to step 7 (out of 12) of the on-going Asia IT&C Project on Multi-Agent Systems (MAS), Social Sciences and Integrated Natural Resource Management (INRM) coordinated by the IRRI-Cirad team based at IRRI Thailand Office. This series of short courses is also supported by IRRI, Cirad, and the Regional Cooperation

Office of the Embassy of France in Bangkok. This time, the working sessions aimed at improving the participants ability to deal with several key methodological issues regarding the use of MAS models to examine social sciences and INRM issues.

Course objectives

The objectives of this new short course were threefold:

- (i) To improve the participants' knowledge on key methodological issues linked to the use of MAS and models in social sciences and INRM,
- (ii) To improve their knowledge on suitable procedures for the validation of social simulation models,
- (iii) To learn new modelling techniques for social simulation.

Summary of the training

The regional training course on MAS, Social Sciences and INRM was held at the Faculty of Science, Chulalongkorn University, Bangkok, on 17-21 March 2003. The local organizing committee created good working conditions and a pleasant atmosphere for the 17 participants coming from 5 different countries.

The focus of the course was on methodological issues regarding the use of MAS and models in social sciences and INRM. Following an overview of the course and an introduction of the course instructor by Dr. F. Bousquet, Pr. Moss described his personal itinerary, from economics to social simulations, and the current activities of his research team. He started to lecture by explaining how to recognize the features of “bad” and of “good” or “better” models. This was illustrated by two examples on climate change and water demand management, followed by an exercise in small groups using the “snow ball discussion” technique. Particularly, the discussion of these examples and small group work led to a comparison between MAS and standard economic modelling approaches.

The next lecture aimed at improving the participants understanding of the procedures for validating MAS models. Both use of statistical signatures and qualitative data were described. A second exercise in small groups focused on: how far extending the validation of MAS models could improve them? On Wednesday morning, Pr. Moss lectured on the role of abstraction for model verification. Of particular interest was a long and illustrated talk on using canonical environments, tools and models to compare models. He continued to teach this subject on Thursday morning by showing the usefulness and comparing the use of grids and tags for designing multi-agent models and for modeling social behavior. This was followed by a long exercise in small groups on designing a multi-agent model based on a real environmental problem proposed by a participant. This last small group exercise was highly-rated by the trainees in their evaluation of the course.

A last lecture aimed at comparing procedural and declarative modeling approaches. It was followed by an introduction and hands-on exercises in the computer laboratory on the Strictly Declarative Modelling Language (SDML) platform used by Pr. Moss' team. The course ended with two detailed presentations made by regular participants from north Thailand and The Philippines on their on-going Ph.D. research on watershed management and on a new project on land-use management of terraced paddies, respectively.

The results of the course evaluation (see details in appendix 3) show that:

- The overall effectiveness of this course was rated as “good” (by some 80% of the participants) or “excellent”. Among the course features, “lectures” and “discussions” (with “meals”!) were the most highly-rated (Figure 5 in appendix 3).
- The 3 objectives of the course were achieved for a large majority of the trainees. But about a fourth of the trainees had problems mastering the procedures for model validation (second objective) and found the lectures difficult to follow.
- Among the different topics of this course, the design of a model in small groups and the presentation and discussion of trainees’ own work were highly-rated for their contents and usefulness to work. But many trainees found the presentation on grids and tags and the demo. on the declarative SDML modelling approach too short and less useful.
- Many topics to be added in this course were suggested by the trainees. Particularly, many underlined again the need for more detailed case studies. But the participants did not suggest any topic to be deleted. In this case, only a longer course could meet their requests.
- The experience of the course instructor, the work in small groups, and the organization of the course and its contents were the main strengths mentioned by the trainees (Figure 6). While the logistical arrangements, the incomplete coverage of the subject matter because of time constraints, and the difficulty of the course were found to be the main weaknesses of this training (Figure 7).

During the week between this short course and the next one at Khon Kaen University, about one half of the participants (from The Philippines, Vietnam, and Thai universities) came to IRRI Thailand office in Bangkok to familiarize themselves with the Cormas simulation platform or to improve the design of their multi-agent models (regular participants in this series of courses) under the guidance of Dr. François Bousquet.

Following two similar short trainings in Europe organized for two Thai participants in 2001 and 2002, thanks to a special support from Cirad, another 2 regular trainees from The Philippines and Bangladesh will soon travel to Montpellier this spring to work for several months with the MAS research team at Cirad and to participate in an international conference on agent-based modelling. A regular participant in this series of short courses is also introducing the contents of this Asia IT&C project to her Japanese colleagues during a 3-month-long stay in Japan. Lastly, a participant has initiated talks with Pr. Moss to apply for a position of visiting scientist for one year in his team at Manchester Metropolitan University.

About the course instructor

Pr. Scott Moss is the Professor of Social Simulation and Director of the Centre for Policy Modelling in the Manchester Metropolitan University. Currently, Pr. Moss is working with a bright, young, multi-disciplinary team which is developing a programming architecture and language called Strictly Declarative Modelling Language (SDML) to model multi-agent interactions in complex institutional and technological environments. SDML is being used to simulating strategic decision-making processes, develop artificially intelligent modelling methods, model complex markets, etc. He is also presently co-coordinator with Rosaria Conte of ABSS, the Agent Based Social Simulation special interest group of AgentLink, the Esprit Network for Excellence for Multi-Agent Systems and joint editor of a special issue of

Computational and Mathematical Organization Theory on representations of cognition. He was also programme chair of MABS2000, the 2nd Multi-Agent Based Simulation workshop federated with ICMAIS-2000 in Boston, July 2000. This training course in Bangkok was his first visit to Southeast Asia.

The participants

17 trainees from five different countries (11 from 6 different universities in Thailand, 3 from the Philippines, and one each from Bangladesh, Vietnam and the Netherlands) attended this one-week training course (see the list of participants in appendix 2).

Beyond several regular participants who attended most of the previous sessions, newcomers joined in the project. Particularly, 3 new computer scientists from Chulalongkorn, Khon Kaen and Cantho Universities participated in this course. This is encouraging as the field of computer science was somewhat under-represented in the previous courses. 4 master students and 2 senior lecturers from the Faculty of Science at Chulalongkorn University also attended all the sessions.

Course outlines

The detailed schedule of the course sessions is presented in appendix 1. This course aimed at improving the participants' ability:

- (i) To recognize the features of “good” models vs “bad” ones: comparisons between MAS and standard economic modelling approaches,
- (ii) To better understand the procedures for validating and verifying MAS models,
- (iii) To realize the usefulness of abstraction using grids, tags, for the design of MAS,
- (iv) To compare procedural and declarative modeling approaches: presentation of the SDML (Strictly Declarative Modelling Language) platform,
- (v) To develop their own personal applications by providing comments and suggestions on on-going or new projects.

Training approach and structure of the course

Pr. Moss' lectures alternated with work in small groups aiming at putting in practice and discussing the contents of the last lecture. The “snowball discussion” technique was applied several time to stimulate participation in the discussions by all trainees and to compare and merge the results achieved by different small groups of 2 persons, then 4, etc. The course evaluation results (see their detailed graphical presentation in appendix 3) show that the trainees liked this general organization of the course. It was well-adapted to the fact that several lectures were found to be difficult to follow by some participants. These trainees got an opportunity to catch-up and get answers to their questions during the small group exercises and discussions following the lectures.

Work in small groups was, as usual, highly-rated by the participants, but it is also rather time-consuming. This is the reason why some topics mentioned in the initial program (like genetic programming and the presentation of several model examples) could not be presented by the

main instructor because of a lack of time. But only 3 participants regreted this choice. The priority given to exercises in small groups also limited the time available for the detailed presentation of models and case studies. A third of the participants said in their evaluation forms that this was a weakness of this course. Little time was also left for hands-on exercises in the computer laboratory. A 2-hour-long session organized on Thursday afternoon was found far too short by a large majority of participants (see figure 4 in appendix 3).

Like in previous courses, 2 presentations by trainees were made in the morning of the last day. Again, the course evaluation shows that the participants appreciate such presentations (see figure 3 in appendix 3):

- Mr. Panomsak Promburom, Ph.D. candidate from the Multiple Cropping Center at Chiang Mai University presented recent progress made on his research regarding MAS, Spatial Modelling and Role-Playing Games for Integrated Watershed Management in Chiang Mai Province, northern Thailand,
- Dr. Rhodora M. Gonzales and Mr. Ariel C. Blanco from the Faculty of Engineering at the University of the Philippines, Diliman Campus, presented their new project entitled: “Geomatics, MAS and 3-D Visualization for Knowledge Interaction in Resource Management: Participatory simulation of ecosystem dynamics with the Ifugaos in the Philippines.

Training materials

Every participant in this course received the following documents:

- (i) A folder entitled “MAS, Social Sciences, and Integrated Natural Resource Management: Methodological Issues” reproducing all the slides used by the course instructor during the week, as well as three reference papers written by him or colleagues in his research team.
- (ii) A CD providing all the files related to the lectures and exercises.

Course evaluation

The results of the course evaluation made by 14 (out of 17) participants are graphically presented in appendix 3.

- (i) Overall effectiveness of the course and achievement of objectives: this course was rated as “good” (by some 80% of the participants) or “excellent”(20%). Among the course features, “lectures” and “discussions” (with “meals”!) were the most highly-rated ones (see figure 5). According to these answers, the 3 objectives of the course were achieved for a large majority of the trainees. But about a fourth of the trainees had problems mastering the procedures for model validation (second objective) as they found the lectures difficult to follow and would have needed more concrete examples and case studies.
- (ii) Topic evaluation: among the ten different topics included in this course, the small group exercise on the design of a model and the presentation and discussion of trainees’ own work were highly-rated for their contents and usefulness to work

(see figure 3). But the presentation on abstraction using grids and tags and the demo. on the declarative SDML modelling approach were found to be too short (especially the computer demo.) and less useful by many participants (especially the members of the core group of trainees already familiar with grids). The trainees suggested to add many topics to this course (see the complete list of their suggestions under figure 3). Particularly, 5 of them expressed the need for more detailed case studies. Several of them mentioned that more time was needed to show the practical application of the procedures for model validation and verification on already completed models or for particular MAS being developed by the group. But as the participants did not suggest to delete any topics (see their few suggestions under figure 4), only a longer course could have met their request for a more extensive coverage of the subject matter.

- (iii) Strengths and weaknesses of the course: a total of 38 strengths were mentioned by 12 participants. The extensive knowledge and long experience of the course instructor was mentioned as a major strength of this training. The work in small groups and the discussions came in second place, followed by the organization and management of the course as well as its contents (see figure 6). Again, the shared experience among regular participants is also seen as a plus by several trainees. The logistical arrangements in the lecture room (particularly the audio and computer facilities) were seen as the main weakness of this training by a fifth of the participants (see figure 7). The incomplete coverage of the subject matter because of lack of time and the difficulty of several topics came in second place. Less important things to be improved were time management and the quality of the documents distributed to the participants.
- (iv) Impact of the course: almost all the participants said that they share their new knowledge with colleagues when they are back at their universities. Exchanges are made with the following types of persons: computer, environmental and social scientists and specialists in community development, development communication, agricultural systems, remote sensing and GIS. Six trainees mentioned colleagues who will be interested to attend similar short courses in the future because of the proposed interdisciplinary and participatory approach to understanding the social system.

Progress made

Pr. Scott Moss was the second social scientist to come from the United Kingdom to run a short course under this Asia IT&C project (the first one was Pr. Nigel Gilbert at Chiang Mai University in February 2002). This week of lectures and small group exercises has reinforced the capacity of the participants in modelling social interactions and relationships between the social agents and their common environment. Compared with previous similar exercises, the presentations made by some regular participants also display clear improvements in the presentation of their respective applications.

New individuals or teams at several Thai universities are now taking part in this project. During the week between this short course and the next one at Khon Kaen University in early April, about one half of strongly motivated participants (from The Philippines, Vietnam, and Thai universities) came to IRRI Thailand office in Bangkok to familiarize themselves with

the Cormas simulation platform or to improve the design of their multi-agent models (regular participants in this series of courses) under the guidance of Dr. François Bousquet.

Following two similar short trainings in Europe organized for two Thai participants in 2001 and 2002, another 2 regular trainees under this project from The Philippines and Bangladesh will travel to Montpellier during spring time to work for several months with the MAS research team at Cirad. Special financial support from Cirad has been secured to make these trips possible. During their stay there, they will also be able to participate in an international conference on agent-based modelling to be held in Montpellier in late April 2003. A regular philipina participant in this series of short courses is currently introducing the contents of this Asia IT&C project to several Japanese colleagues during a 3-month-long stay in Japan. Lastly, a new participant in this course has initiated talks with Pr. Moss to apply for a position of visiting scientist for one year in his team at Manchester Metropolitan University.

The next meeting of this network will take place at Khon Kaen University during 29 March-4 April and the leading course instructor will be Dr. Wander Jager, another social scientist from the University of Groningen in The Netherlands. Later in the year, another short course is planned in October 2003 at Chiang Mai University. This last 1-week course for 2003 will focus on MAS and participatory approaches and will be led by Pr. Niels Röling from Wageningen University and Research Center, also in The Netherlands.

It will be followed by a shorter workshop among the network members developing models to help understanding the resilience of agroecosystems, effects of different options for managing resources, and the role of social organizations for regulating landscape dynamics. This workshop will be a key phase in the preparation of a first publication of this regional network presenting a series of some 10 case studies dealing with a range of concrete INRM problems in the region. The preparation of this collective book by MAS for INRM practitioners in Southeast Asia was discussed at the end of this training course and the production of extended abstracts of the papers has been initiated.

Appendices

1. Course schedule.
2. List of participants.
3. Results of the course evaluation by the participants.

Appendix 1. Schedule of the International Training on “Multi-Agent Systems, Social Sciences, and INRM: Methodological Issues” at the Faculty of Science, Chulalongkorn University, Bangkok, 17-21 March 2003.

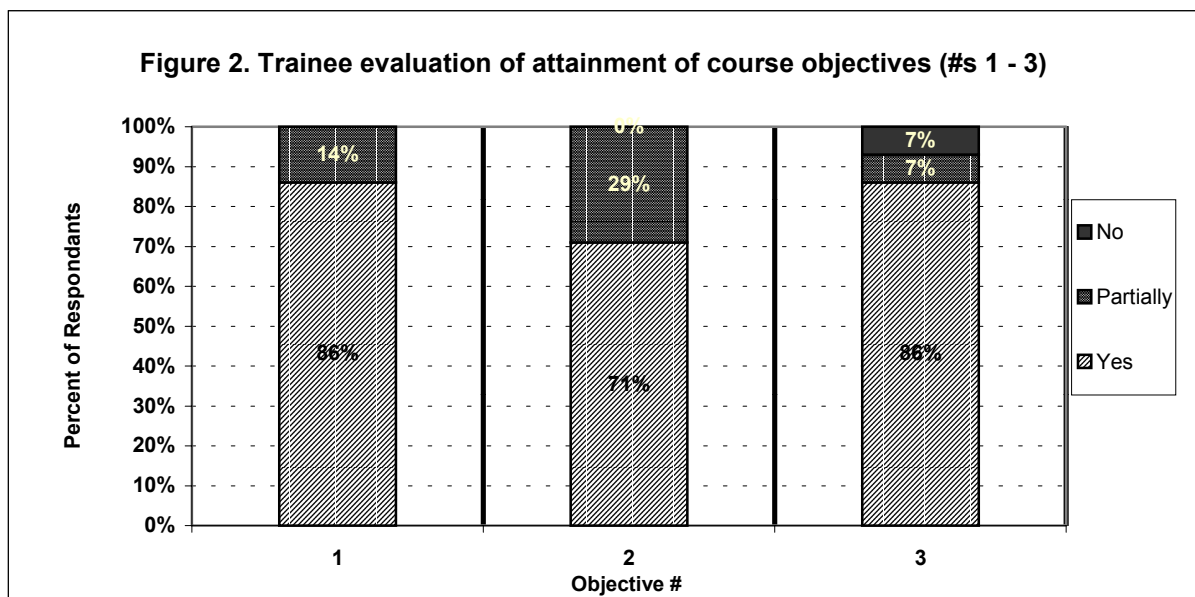
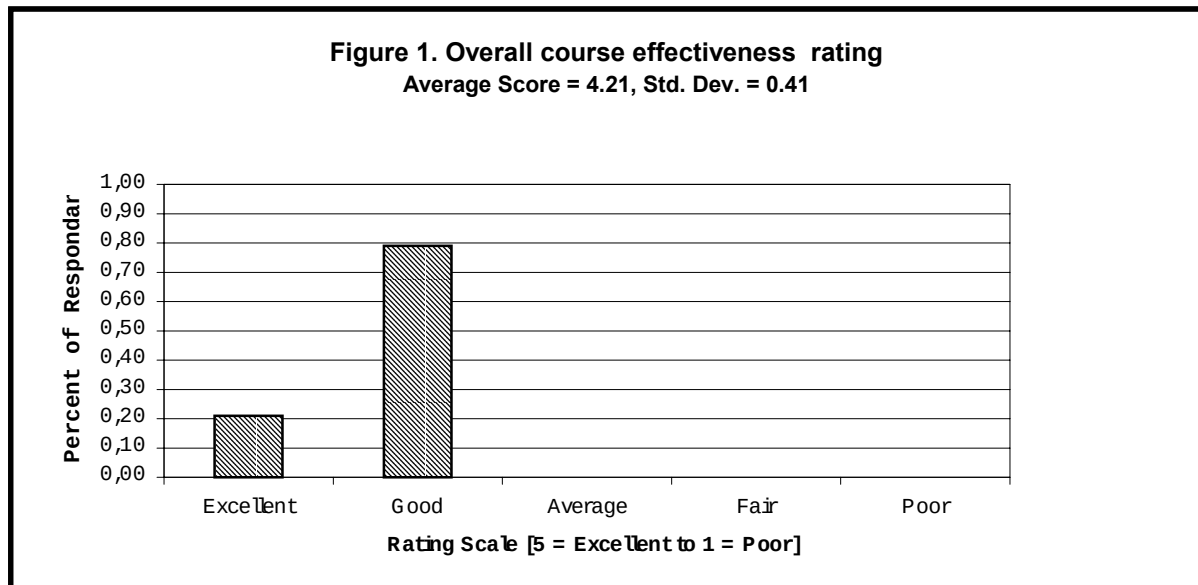
	Monday	Tuesday	Wednesday	Thursday	Friday
8h30-9h00	Registration	Validation: statistical signatures and qualitative data	Verification: the role of abstraction	Abstraction: using grids and tags	Presentations by participants and discussion: 1. North Thailand 2. Philippines Cordillera
9h00-10h30	Opening ceremony and briefing by Chair of organizing committee				
10h30-12h00	Overview of the course by the main course instructor	Group work 2: extending validation to improve good or bad models	Verification: canonical models	Group work 3 on designing a MAS on a real environmental problem.	Evaluation and closing ceremony
13h30-14h10	Lecture on “bad” models	Validation: case study of the water demand model	Free discussions and small meetings	Modelling approaches: procedural and declarative	Meeting of trainees developing applications: preparation of October workshop.
14h20-15h00	Lecture on “good/better” models			Demo. on declarative modelling in SDML	
15h30-17h00	Group work 1 on “good” and “bad” models	Group work 3: validation for applied MAS models			

Appendix 2. List of participants in the training course on Multi-Agent Systems, Social Sciences, and INRM at the Faculty of Science, Chulalongkorn University, Bangkok, 17-21 March 2003.

Dr. Nantana Gajasen	Faculty of Science, Chulalongkorn University	gnantana@chula.ac.th
Dr. Wina Mekvichai	Faculty of Science, Chulalongkorn University	wina@sc.chula.ac.th
Ms. Sirisuk Aratrakorn	Faculty of Science, Chulalongkorn University	arrathrakorn@yahoo.com
Mr. Pongchai Dumrongrojwatthana	Faculty of Science, Chulalongkorn University	dpongchai@chula.ac.th
Mr. Kobchai Worraphimphong	Faculty of Science, Chulalongkorn University	wkobchai@hotmail.com
Mr. Chanchai Chaisukkosol	Faculty of Political Science, Chulalongkorn University	chairainbow@chula.ac.th
Mr. Panomsak Promburom	Multiple Cropping Center, Chiang Mai University	agopprmb@chiangmai.ac.th
Ms. Nongluck Suphanchaimart	Faculty of Agriculture, Khon Kaen University	nongluck@kku.ac.th
Dr. Somjit Arch-int	Faculty of Science, Khon Kaen University	somjit@kku.ac.th
Mr. Warong Naivinit	Faculty of Agriculture, Ubon Ratchathani University	warong@agri.ubu.ac.th
Dr. Suwannee Adsavakulchai	King Mongkut's University of Technology, Thonburi	suwannee@jgsee.kmutt.ac.th
Mr. Morshed Anwar	STAR Program, Asian Institute of Technology, Rangsit	iamhiraj@yahoo.com
Mr. Stanislas Boissau	SAM Regional Program, VASI,	sboissau@fpt.vn

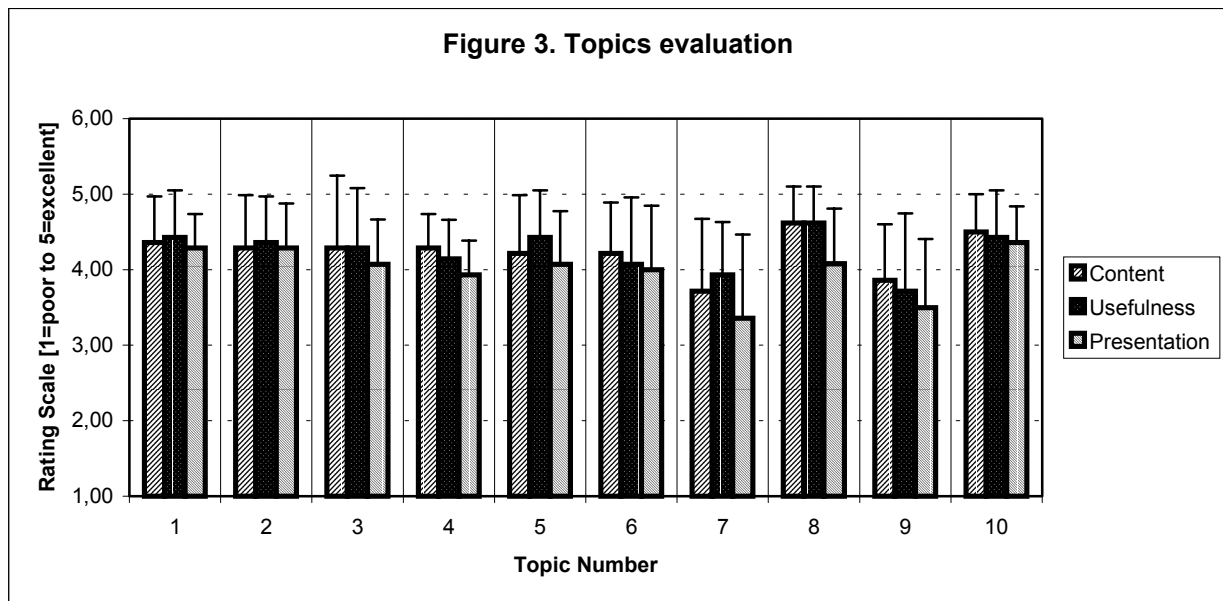
	Hanoi, Vietnam	
Mr. Vinh Nhi Gia Nguyen	College of Science, Cantho University, Vietnam	aviat@yahoo.com
Dr. Rhodora M. Gonzalez	College of Engineering, University of the Philippines, Manila	rhogon@yahoo.com
Mr. Ariel C. Blanco	College of Engineering, Univ. of the Philippines, Manila	ayeh@yahoo.com
Ms. Princess Alma B. Ani	Eco-Informatics Lab, Univ. of the Philippines, Los Banos	princessanidala@yahoo.com

Appendix 3. Results of the course evaluation by the participants: Multi-Agent Systems, Social Sciences, and INRM at the Faculty of Science, Chulalongkorn University, Bangkok, 17-21 March 2003.



Course objectives:

1. To improve the participants' knowledge on key methodological issues linked to the use of multi-agents systems (MAS) and models in social sciences and INRM,
2. To improve their knowledge on suitable procedures for the validation of social simulation models,
3. To learn new modelling techniques for social simulation.



List of topics

1. Overview of the course
2. "Bad" and "good" models
3. Group work 1 on "bad" and "good" models
4. Validation: statistical procedures and qualitative data
5. Group work 2 on how to improve models by extending validation
6. Verification: role of abstraction and canonical models
7. Abstraction: grids and tags
8. Group work 3 on designing an applied MAS model (mining village)
9. Demo on procedural and declarative SDML modelling approaches
10. Presentation of models by participants and discussion

Topics to be added

- More detailed examples under each topic and case studies (5 participants)
- More examples on applying validation procedures (2)
- Genetic programming (2)
- SDML modelling practice (2)
- Logical method to link abstraction and design
- Validation and verification techniques applied in existing and on-going MAS for INRM applications across range of disciplines
- Validation and declarative modelling
- More theoretical background related to some topics like canonical models
- Running a simulation of a model in SDML
- Other modelling procedures related to MAS (a new participant?)
- How to apply MAS models (a new participant)